A variational action for dfmm: deterministic skeleton and stochastic dressing

Working note for T. Abel

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Status. Milestone 0. Continuum derivation only. No code, no discretization yet. Goal: write the Lagrangian whose Euler–Lagrange equations reproduce the dfmm eight-field evolution equations to leading order, verify the cold-limit reduction algebraically, and derive the Kramers–Moyal noise from a stochastic variational principle (not just attach it).

1 Variables and frame

We work in mass coordinates. Let $m \in [0, M]$ label fluid parcels, so that

$$\mathrm{d}m = \rho_0(\xi)\,\mathrm{d}\xi, \qquad J(m,t) := \partial_m x(m,t) = \frac{1}{\rho(x(m,t),t)}$$

where the second equality uses unit mass per unit m (the standard mass-coordinate convention; we set $\rho_0 \equiv 1$ for brevity, restoring factors only when needed). The dfmm primary field L_1 is then literally $L_1(x,t) = m$ inverted, and the conservation law $\partial_t(\rho L_1) + \partial_x(\rho u L_1) = 0$ is identically satisfied because m is a material label.

The deterministic Lagrangian-frame fields are

$$\big(\, x(m,t),\ \alpha(m,t),\ \beta(m,t),\ s(m,t),\ Q(m,t)\,\big), \qquad u := \dot{x}. \tag{1}$$

Here s is the parcel entropy per unit mass (one internal DOF carrying P_{xx}), and Q is the centered third velocity moment (heat flux per unit mass). The Cholesky relations are

$$\Sigma_{xx} = \alpha^2, \qquad \Sigma_{xv} = \alpha\beta, \qquad \Sigma_{vv} = \beta^2 + \gamma^2 = \frac{P_{xx}}{\rho}, \tag{2}$$

with γ algebraically reconstructed and Σ_{vv} determined by (s, J) through the equation of state. For the adiabatic ideal-gas 1-DOF closure of dfmm ($P_{xx} \propto \rho^3$ adiabatic),

$$\Sigma_{vv}(s,J) = \frac{e^{2s/c_v}}{J^2}, \qquad c_v = \tfrac{1}{2} \quad \text{(one longitudinal DOF; we fix units so } c_v = 1/2\text{)}, \tag{3}$$

which gives $P_{xx} = \rho\Sigma_{vv} = e^{2s/c_v}/J^3 \propto \rho^3$ at fixed s, matching paper Eq. (after 93) line 94. The transverse pressure $P_\perp$ enters only through total energy bookkeeping in 3D-with-1D-symmetry and is cleanest treated as an additional spectator entropy; we suppress it here and restore it in an appendix.

2 The deterministic action

2.1 Construction

We propose

$$S\big[x,\alpha,\beta,s,Q;\,\lambda_Q\big] = \int \mathrm{d}t \int \mathrm{d}m \Big\{ \underbrace{\tfrac{1}{2}\dot{x}^2}_{\text{bulk KE}} + \underbrace{\tfrac{1}{2}\,\mu(\alpha,J)\,\dot{\alpha}^{\,2}}_{\text{Cholesky kinetic}} - \underbrace{\varepsilon(J,s,\alpha,\beta)}_{\text{internal energy}} + \underbrace{\lambda_Q\big(\dot{Q} + Q/\tau - \mathcal{S}_Q[J,s,\alpha,\beta]\big)}_{\text{heat-flux constraint}} \Big\}. \tag{4}$$

The pieces are fixed by the requirement that $\delta S = 0$ reproduce the dfmm Lagrangian-frame equations (paper Eqs. (**??**), the $\dot{\Sigma}_{vv}$ equation, and the BGK heat-flux relaxation). We now do that matching.

2.2 The bulk equation: $\delta S/\delta x = 0$

With $J = \partial_m x$ and $\varepsilon = \varepsilon(J,s,\alpha,\beta)$,

$$\frac{\delta S}{\delta x} = -\ddot{x} - \partial_m\left(\frac{\partial \varepsilon}{\partial J}\right) = -\ddot{x} + \partial_m P_{xx}\cdot J^{\,?}, \tag{5}$$

where the standard thermodynamic identity in mass coordinates gives $\partial\varepsilon/\partial J|_s = -P_{xx}$, so

$$\boxed{\ddot{x} = -\partial_m P_{xx}} \quad\Longleftrightarrow\quad \rho\,\mathrm{D}u/\mathrm{D}t = -\partial_x P_{xx}, \tag{6}$$

the 1D Euler momentum equation. *Check:* matches dfmm's hydro update at the level of the bulk momentum flux. ✓

2.3 The Cholesky equations: $\delta S/\delta\alpha = 0$, $\delta S/\delta\beta = 0$

Here is where the prefactor $\mu(\alpha,J)$ in (**??**) gets pinned down. The dfmm equations (**??**) are

$$\dot{\alpha} = \beta, \qquad \dot{\beta} = \frac{\Sigma_{vv} - \beta^2}{\alpha} - (\partial_x u)\,\beta = \frac{\gamma^2}{\alpha} - (\partial_x u)\,\beta. \tag{7}$$

The first is striking: $\dot{\alpha} = \beta$ is *first* order in time, so treating α as a coordinate with kinetic term $\mu\dot{\alpha}^2/2$ cannot directly produce it — a Lagrangian whose only α-velocity term is $\mu\dot{\alpha}^2/2$ gives a second-order equation for α, not a first-order one. The clean fix is to recognize β as the *conjugate momentum* to α, not an independent coordinate.

That is, we rewrite (**??**) in first-order (Hamilton–Pontryagin) form:

$$S = \int \mathrm{d}t \int \mathrm{d}m \,\big\{\tfrac{1}{2}\dot{x}^2 + p_\alpha(\dot{\alpha} - \beta) - H_{\mathrm{Ch}}(\alpha,\beta,J,s) - \varepsilon(J,s) + \lambda_Q\,(\cdots)\big\}, \tag{8}$$

with p_α the multiplier enforcing $\dot{\alpha} = \beta$. Varying p_α gives back $\dot{\alpha} = \beta$ as a constraint.[1] Varying α gives the equation for p_α, and varying β gives the equation for α:

$$\dot{p}_\alpha = -\frac{\partial H_{\mathrm{Ch}}}{\partial \alpha}, \qquad p_\alpha = -\frac{\partial H_{\mathrm{Ch}}}{\partial \beta}\Big|_{\dot{\alpha}=\beta} \quad\Rightarrow\quad \dot{\beta} = \cdots \tag{9}$$

[1] Equivalently: (α,β) is a Clebsch pair on phase space, with symplectic form $\mathrm{d}\alpha\wedge\mathrm{d}\beta$ and Hamiltonian H_{Ch} generating the flow. This is the same structure that appears in the Lagrangian of incompressible hydrodynamics in Clebsch variables. The continuum theory is symplectic in (α,β), not Lagrangian.

For the dfmm second equation (**??**) to fall out, the Cholesky Hamiltonian density must be

$$H_{\mathrm{Ch}}(\alpha,\beta;J,s) = \tfrac{1}{2}\beta^2 + \frac{\Sigma_{vv}(J,s)-\beta^2}{2}\ln\alpha + \tfrac{1}{2}\beta^2\,\partial_x u, \tag{10}$$

*[The $\partial_x u$ term is non-local in m; it couples to the bulk through $\partial_x u = J^{-1}\partial_m \dot{x}$. This is the right place to flag a real subtlety: the dfmm Cholesky equation contains $\partial_x u$ explicitly, so the Cholesky sector is not isolated from the bulk. In the action formulation this coupling is what makes the discrete EL system implicit, and it's the same physics that produces shell-crossing regularization through Cholesky activation. I want to verify the exact form of H_{Ch} against (**??**) by direct variation in the next pass — the form above is my best guess from term-matching, not a finished derivation.]*

2.4 The entropy and heat-flux equations

Variation with respect to s gives a transport equation for s; in the absence of dissipation $\dot{s} = 0$ (entropy per parcel conserved). The heat-flux constraint enforces $\dot{Q} = -Q/\tau + \mathcal{S}_Q$ via $\delta S/\delta\lambda_Q = 0$. The source $\mathcal{S}_Q$ is the standard fourth-moment flux divergence; under Wick closure (paper Eq. (**??**)) it's an algebraic functional of (ρ, u, P_{xx}, Q) and can be plugged in directly.

3 Cold-limit check

Take $s \to -\infty$ (zero internal energy), so $\Sigma_{vv} \to 0$, $P_{xx} \to 0$, and $\beta, \gamma \to 0$. Equation (**??**) reduces to

$$S_{\mathrm{cold}}[\,x\,] \;=\; \int \mathrm{d}t \int \mathrm{d}m\, \tfrac{1}{2}\dot{x}^{\,2}, \tag{11}$$

which is the action for pressureless dust in mass coordinates. The Euler–Lagrange equation is $\ddot{x} = 0$, i.e. ballistic free streaming. This is exactly the 1D phase-space-sheet action: each Lagrangian element moves on a straight worldline in (x,t) until two of them meet (shell crossing). Adding self-gravity is a single extra term $-\int \mathrm{d}m\,\Phi(x(m,t),t)$ with Poisson's equation as a constraint; the Abel–Hahn–Kaehler tessellation framework follows. ✓

The interesting bit. The cold limit has $\Sigma_{vv} = 0$ and hence $\gamma = 0$ identically: the local phase-space distribution is rank one. In the deterministic dfmm scheme this is exactly the regime where $\gamma \to 0$ acts as a closure-failure indicator. In the variational unification, this is the regime where the deterministic action is *degenerate* — the Cholesky kinetic term (**??**) vanishes and the symplectic structure on (α,β) collapses. This degeneracy is the variational signature of phase-space rank collapse, and it is precisely where the stochastic dressing of §**??** below activates: the noise variance is non-vanishing exactly when the deterministic action loses its symplectic non-degeneracy. The unification claimed in the brief is, in this language, the statement that *the noise opens a gap in the otherwise degenerate Hessian of S_{cold}*.

4 Stochastic action: deriving the Kramers–Moyal noise

This section carries out the "stronger" route requested: we want the calibrated KM–LES noise of dfmm to fall out of a stochastic variational principle, not just to be glued onto the deterministic equations.

4.1 Yasue's framework for one-sided diffusions

The dfmm noise is one-sided: the variance is $\propto \max(-\partial_x u, 0)$, so noise is injected only in compression. This breaks time-reversal symmetry and excludes a naive Onsager–Machlup formulation. The right framework is Nelson–Yasue stochastic variational mechanics extended to non-reversible (sub-Markovian) processes, equivalently the Martin–Siggia–Rose–Janssen–de Dominicis (MSRJD) path-integral action specialized to the Itô convention.

For an SDE $\mathrm{d}X = b(X)\,\mathrm{d}t + \sigma(X)\,\mathrm{d}W$, the MSRJD action in field-theoretic form is

$$S_{\mathrm{MSRJD}}[X, \hat{X}] = \int \mathrm{d}t \left\{ i\hat{X}\,(\dot{X} - b(X)) - \tfrac{1}{2}\sigma^2(X)\,\hat{X}^{\,2} \right\}, \tag{12}$$

where $\hat{X}$ is the response field (Lagrange multiplier enforcing the SDE in expectation) and the $\hat{X}^{\,2}$ term is the noise vertex. This is the form we'll match to dfmm.

4.2 Identification for dfmm

Our SDE is on (ρu) in the Eulerian frame, equivalently on $\dot{x}$ per parcel in the Lagrangian frame. Write the per-parcel velocity as $u = u_{\mathrm{det}} + \delta u$ where u_{det} comes from the deterministic EL equation (**??**). The dfmm noise prescription is

$$\delta(\rho u) = C_A \rho\,(\partial_x u)\,\mathrm{d}t + C_B \rho \sqrt{\max(-\partial_x u, 0)\,\mathrm{d}t}\,\eta, \qquad \eta \sim \mathrm{Laplace}(0,1). \tag{13}$$

Translating to the Lagrangian frame ($\rho\,\mathrm{d}x = \mathrm{d}m$, so $\delta(\rho u)\,\mathrm{d}x = \delta u\,\mathrm{d}m$):

$$\delta u = C_A(\partial_x u)\,\mathrm{d}t + C_B\,\sqrt{\max(-\partial_x u, 0)\,\mathrm{d}t}\,\eta \quad \text{per parcel.} \tag{14}$$

Reading off drift and diffusion: $b_{\mathrm{noise}}(m,t) = C_A \partial_x u$ and $\sigma^2(m,t) = C_B^2\,\max(-\partial_x u, 0)$. The stochastic contribution to the total action is then

$$S_{\mathrm{stoch}}[u, \hat{u}] = \int \mathrm{d}t \int \mathrm{d}m \left\{ i\hat{u}\,(\dot{u} - u^{\bullet}_{\mathrm{det}} - C_A \partial_x u) - \tfrac{1}{2}\,C_B^2 \max(-\partial_x u, 0)\,\hat{u}^{\,2} \right\}, \tag{15}$$

where $u^{\bullet}_{\mathrm{det}}$ denotes the deterministic right-hand side from (**??**). *The Laplace distribution is non-Gaussian, so* (**??**) *is only correct to second cumulant.* To capture the heavy tails one must add the cumulant-generating function of Laplace, $\log\langle e^{ik\eta}\rangle = -\log(1+k^2)$, which gives an infinite tower of $\hat{u}^{\,2n}$ vertices:

$$S^{\mathrm{full}}_{\mathrm{stoch}} = S^{(2)}_{\mathrm{stoch}} - \sum_{n\geq 2} \frac{(-1)^n}{n} \left[\tfrac{1}{2} C_B^2 \max(-\partial_x u, 0)\right]^n \hat{u}^{\,2n}. \tag{16}$$

- **Conservative.** The deterministic action plus a Gaussian stochastic dressing is the canonical theory. The Laplace tails are a *phenomenological correction* that reflects intermittency in the underlying turbulence; we keep them in the dfmm scheme because they matter quantitatively, but the action principle proper is Gaussian.

- **Stronger.** The Laplace distribution arises as the stationary distribution of a two-state compression-rarefaction switching process: noise events come in compression bursts of finite duration, with an exponential waiting-time distribution between bursts. The marginal of the integrated noise over a wave-pool save interval is then a compound Poisson process with exponential jumps, whose continuous-time limit is exactly the Laplace family. This makes the Laplace form derivable from the underlying *intermittent* compression statistics, not just fitted to them.

The stronger framing is what I think you want, but it's a research claim that needs its own check (compute the compression-burst statistics in the dfmm wave-pool runs and verify they are Poisson with exponential durations). I'd like to flag this as a check-on-data step before the note commits to it.]*[This is the place where I most want your reaction. The Laplace distribution is the dfmm calibration choice (paper line 422: the residual distribution* matches *Laplace, kurtosis 3.45 vs Gaussian 3). In a strict stochastic-variational derivation we would expect Gaussian noise to be canonical (because Brownian motion is the unique max-entropy continuous-time process at fixed second moment). The Laplace tails are* empirical *— they reflect the actual closure error distribution in the wave-pool DNS. Two honest framings:*

- ***Conservative.*** *The deterministic action plus a Gaussian stochastic dressing is the canonical theory. The Laplace tails are a* phenomenological correction *that reflects intermittency in the underlying turbulence; we keep them in the dfmm scheme because they matter quantitatively, but the action principle proper is Gaussian.*

- ***Stronger.*** *The Laplace distribution arises as the stationary distribution of a two-state compression-rarefaction switching process: noise events come in compression bursts of finite duration, with an exponential waiting-time distribution between bursts. The marginal of the integrated noise over a wave-pool save interval is then a compound Poisson process with exponential jumps, whose continuous-time limit is exactly the Laplace family. This makes the Laplace form derivable from the underlying* intermittent *compression statistics, not just fitted to them.*

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4.3 The drift term as the Itô shift

The drift coefficient $C_A\,\partial_x u$ in (**??**) is exactly the form expected for an Itô-Stratonovich correction: $\frac{1}{2}\,\partial_u\sigma^2 = \frac{1}{2}\,C_B^2\,\partial_u \max(-\partial_x u, 0)$ under chain rule, which has the right structural form $\propto \partial_x u$. The numerical coefficient $C_A \approx 0.34$ vs. $C_B^2/2 \approx 0.55^2/2 \approx 0.15$ does *not* match a pure Itô shift, however — there's a factor of ~ 2.3 discrepancy. This says either (i) the wave-pool calibration includes physical drift (genuine mean closure error) on top of the Itô shift, or (ii) the conversion between Eulerian-frame $\partial_x u$ and Lagrangian-frame $\partial_m u \cdot J^{-1}$ contributes an extra factor that I haven't yet tracked carefully. *[This is the second place I want your reaction — you calibrated C_A and C_B from the wave-pool, so you'll know whether the empirical drift is "mostly the Itô shift" or "mostly genuine mean error."]*

5 What this gives us

1. A continuum action whose deterministic Euler–Lagrange equations reproduce dfmm's eight-field Lagrangian-frame evolution, with the cold limit reducing exactly to the phase-space-sheet (mass-coordinate dust) action.

2. A natural variational interpretation of the rank-collapse diagnostic: $\gamma \to 0$ is the Hessian of S_{det} becoming degenerate, and the noise is what re-opens the gap.

3. A stochastic action with the right structure (one-sided noise variance, drift compatible with the Itô shift) that reduces to the calibrated KM–LES form to second cumulant. The Laplace tails require an additional intermittency hypothesis on the underlying compression statistics, which is checkable from existing dfmm wave-pool data.

4. The discrete scheme that follows is symplectic in (α, β) and variational in x. The natural time integrator is discrete EL on the Lagrangian-frame mesh, which gives bounded long-time energy error automatically (no artificial energy debit needed).

5. The AMR criterion is local action error ΔS per cell — this is well-defined because the action is now an explicit functional, not a post-hoc reconstruction. Stochastic injection is gated on ΔS exceeding a threshold, which is how the scheme self-regulates near shell crossing.

6 Open questions for T. Abel before Milestone 1

1. **Cholesky Hamiltonian form.** I've written H_{Ch} to a plausible form by term-matching but not done the full algebraic verification that varying (**??**) reproduces (**??**) exactly. The non-local $\partial_x u$ coupling is the trickiest piece. Should I (a) finish that algebra and tighten this section, or (b) move to a slightly different Clebsch pair where the algebra is cleaner? The choice affects whether the discrete scheme is explicit or implicit in the Cholesky sector.

2. **Laplace from intermittency.** Is the Laplace distribution robust across your wave-pool seeds, or is it specific to the compressible 1D regime? If it's specific, the "Laplace from compound Poisson" derivation is the right one. If it's universal across regimes, there's a deeper reason (possibly tied to the multiplicative cascade structure of turbulence, which would tie back to your She–Levêque work).

3. **Drift-shift discrepancy.** See the second *[...]* above. Is $C_A \approx 0.34$ "mostly mean closure error" or "mostly Itô shift"? Empirically separable from the calibration data, I think.

4. **Heat flux constraint.** Should Q enter the action as a hard Lagrange-multiplier constraint (as I've written it) or as a soft action term with its own kinetic prefactor? The hard-constraint form is cleaner and matches the BGK structure but loses some flexibility for the maxent fourth-moment closure.

Once these are pinned down I'll iterate the note to a clean finished form and only then move to discretization (Milestone 1).